WORKED SOLUTION

STEP III 1991, Question 12

Mechanics • Projectile motion

STEP III 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q12

The key idea

The particle is tied to the ellipse, so its position is completely described by the single parameter θ. Differentiate the position vector twice and project the acceleration onto the inward unit normal: the $\ddot{\theta}$ terms drop out, because they point along the tangent. For the second part, the particle leaves the surface when the normal reaction falls to zero, and the tube is smooth, so energy conservation gives the speed at every point.

Solution

The normal acceleration

We model the particle as a point moving on a smooth curve, so the only forces are its weight mg downwards and a normal reaction R from the tube.

The position is $\mathbf{r} = (a\cos\theta,\ b\sin\theta)$. Differentiating with respect to time, using the chain rule,

$$\dot{x} = -a\sin\theta\,\dot{\theta}, \qquad \dot{y} = b\cos\theta\,\dot{\theta},$$
$$\ddot{x} = -a\sin\theta\,\ddot{\theta} - a\cos\theta\,\dot{\theta}^2, \qquad \ddot{y} = b\cos\theta\,\ddot{\theta} - b\sin\theta\,\dot{\theta}^2.$$

The tangent direction is $(-a\sin\theta,\ b\cos\theta)$, so a normal direction is $(b\cos\theta,\ a\sin\theta)$ (their dot product is 0). At $\theta = 0$ this is $(b, 0)$, which points out of the ellipse, so the inward unit normal is

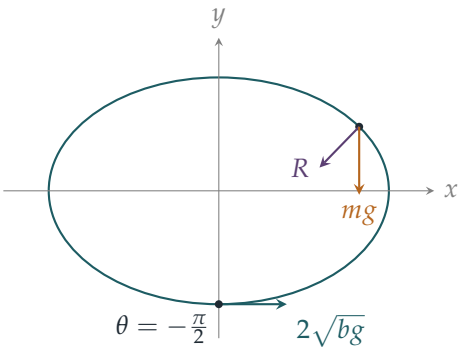
$$\mathbf{n} = -\frac{(b\cos\theta,\ a\sin\theta)}{D}, \qquad D = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}.$$

The acceleration along the inward normal is therefore

$$\ddot{\mathbf{r}}\cdot\mathbf{n} = -\frac{1}{D}\Big[b\cos\theta(-a\sin\theta\,\ddot{\theta} - a\cos\theta\,\dot{\theta}^2) + a\sin\theta(b\cos\theta\,\ddot{\theta} - b\sin\theta\,\dot{\theta}^2)\Big]$$
$$= -\frac{1}{D}\Big[-ab\,\dot{\theta}^2(\cos^2\theta + \sin^2\theta)\Big] = \frac{ab\,\dot{\theta}^2}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}},$$

since the two $\ddot{\theta}$ terms cancel. This is the required result.

Leaving the surface



The lowest point has $\theta = -\frac{\pi}{2}$, where $y = -b$. The tube is smooth, so the reaction does no work and energy is conserved. With speed v at the point with parameter θ,

$$\tfrac{1}{2}mv^2 + mgb\sin\theta = \tfrac{1}{2}m(4bg) - mgb \quad \Longrightarrow \quad v^2 = 2gb(1-\sin\theta).$$

From the first part, $v^2 = \dot{x}^2 + \dot{y}^2 = D^2\dot{\theta}^2$, so $\dot{\theta}^2 = v^2/D^2$.

Now resolve along the inward normal $\mathbf{n}$. The reaction R acts along $\mathbf{n}$ (the particle is on the inside, so the tube pushes inwards and $R \geq 0$ while there is contact). The weight is $(0, -mg)$, whose component along $\mathbf{n}$ is $(0,-mg)\cdot\mathbf{n} = mga\sin\theta/D$. Newton's second law gives

$$R + \frac{mga\sin\theta}{D} = \frac{m\,ab\,\dot{\theta}^2}{D} = \frac{m\,ab\,v^2}{D^3}, \qquad \text{so} \qquad R = \frac{ma}{D^3}\left(bv^2 - g\sin\theta\, D^2\right).$$

Ask yourself Why is the weight's component $+mga\sin\theta/D$, and not with a minus sign?

In the upper half ($\sin\theta > 0$) the inward normal has a downward component, so gravity helps to pull the particle inwards there, just as at the top of a vertical circle. Both R and this part of the weight supply the centripetal acceleration.

Now put $a = \frac{3}{2}b$ and $v^2 = 2gb(1-\sin\theta)$, and write $s = \sin\theta$. Then

$$D^2 = \tfrac{9}{4}b^2s^2 + b^2(1-s^2) = b^2(1+\tfrac{5}{4}s^2),$$

so

$$bv^2 - g\sin\theta\, D^2 = gb^2\left[2(1-s) - s(1+\tfrac{5}{4}s^2)\right] = -\tfrac{1}{4}gb^2(5s^3+12s-8).$$

Since $ma/D^3 > 0$, the sign of R is the sign of $-(5s^3+12s-8)$.

The particle leaves the surface where R first becomes zero. The function $p(s) = 5s^3+12s-8$ is strictly increasing, because $p'(s) = 15s^2+12 > 0$, and $p(-1) = -25 < 0$,

$p(1) = 9 > 0$. So p has exactly one root in $(-1, 1)$. As the particle rises from the lowest point, s increases from -1, and $R > 0$ until s reaches that root. Also $v^2 = 2gb(1 - s) > 0$ for all $s < 1$, so the particle is still moving up the tube when it gets there. Hence R first vanishes, and the particle leaves the surface, at the point where

$$5\sin^3\theta + 12\sin\theta - 8 = 0,$$

as required. Numerically, $\sin\theta \approx 0.584$, so it leaves a little above the level of the axis.

Checks

- If $a = b = r$ the normal acceleration becomes $r\dot{\theta}^2 = v^2/r$, the familiar circular result.
- At the lowest point, $R = m\left(\frac{bv^2}{a^2} + g\right)$ with $v^2 = 4bg$, which is positive (the radius of curvature there is a^2/b), and $p(-1) = -25 < 0$ agrees.
- With $b = 1$, $a = 1.5$ and $g = 9.81$ we integrated the equation of motion in θ step by step from the lowest point. The reaction first vanished at $\sin\theta \approx 0.58379$, matching the root 0.58377 of the cubic ($\theta \approx 35.7°$).

Common slips

- Treating θ as the polar angle. The point $(a\cos\theta, b\sin\theta)$ is not at angle θ from the x-axis unless $a = b$, and the normal is not along the radius.
- Using v^2/b or v^2/a for the normal acceleration. On an ellipse the radius of curvature changes from point to point, which is exactly what the first part computes.
- Getting the sign of the weight's normal component wrong, which turns the cubic into $5s^3 - 4s + \ldots$ and gives no sensible root.

Beyond STEP

The first part is a special case of the formula v^2/ρ for normal acceleration, where ρ is the radius of curvature. Indeed, $\rho = D^3/(ab)$ for this ellipse. Splitting acceleration into tangential and normal parts (the Frenet frame) is standard in first-year mechanics and differential geometry. Treating the constraint with one coordinate θ leads naturally to Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers constrained motion and polar and intrinsic coordinates well.

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